

Discrete Mathematics

Graph Theory

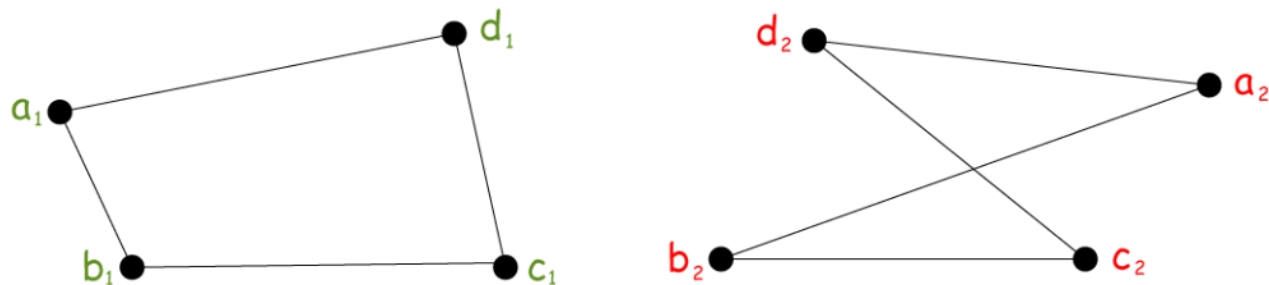


Isomorphism of Graphs

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 for all $a, b \in V_1$.

Such a function f is called an **isomorphism**.

Two graphs are **isomorphic** if they represent the **same relationships** between elements, even though they might look different visually.



Note

Two simple graphs that are not isomorphic are called **non-isomorphic**.

1. Isomorphic simple graphs must have the same number of vertices, because there is a one-to-one correspondence between the sets of vertices of the graphs.

2. Isomorphic simple graphs also must have the same number of edges, because the one-to-one correspondence between vertices establishes a corresponding relation between edges.

3. The degree of the vertices in isomorphic simple graphs must be the same.

Define – Isomorphism (method)

Two graphs G_1 and G_2 are isomorphic if there is a function

$$f : V(G_1) \rightarrow V(G_2)$$

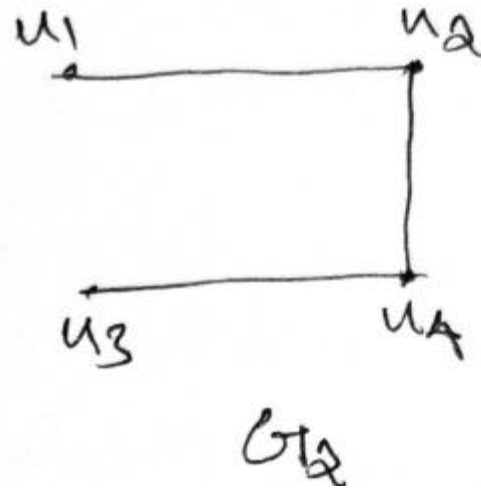
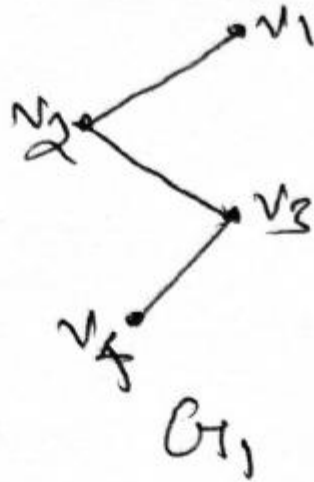
from the vertices of G_1 to the vertices of G_2 such that

- ✓ i) f is one-to-one
- ✓ ii) f is onto
- ✓ iii) For each pair of vertices u and v of G_1 ,

$$(u, v) \in E(G_1) \Leftrightarrow (f(u), f(v)) \in E(G_2)$$

Any function f with the above three properties is called an **isomorphism** from G_1 to G_2 .

1 Show that the two graphs are isomorphic



Soln

Given G_1 and G_2 having four vertices and three edges
 $V(G_1) = \{v_1, v_2, v_3, v_4\}$ $V(G_2) = \{u_1, u_2, u_3, u_4\}$
 $E(G_1) = \{(v_1, v_2), (v_2, v_3), (v_2, v_4)\}$, $E(G_2) = \{(u_1, u_2), (u_2, u_4), (u_4, u_3)\}$

let us define a function $f: V(\sigma_1) \rightarrow V(\sigma_2)$ as

$$f(v_1) = u_1, \quad f(v_2) = u_2, \quad f(v_3) = u_4, \quad f(v_4) = u_3$$

clearly f is one-one and onto.

also [To check vertex preserves]

$$(v_1, v_2) \in E(\sigma_1) \quad \text{then} \quad (f(v_1), f(v_2)) = (u_1, u_2) \in E(\sigma_2)$$

$$(v_2, v_3) \in E(\sigma_1) \quad \text{then} \quad (f(v_2), f(v_3)) = (u_2, u_4) \in E(\sigma_2)$$

$$(v_3, v_4) \in E(\sigma_1) \quad \text{then} \quad (f(v_3), f(v_4)) = (u_4, u_3) \in E(\sigma_2)$$

$$\text{||} (v_1, v_4) \notin E(\sigma_1)$$

$$\text{||} (f(v_1), f(v_4)) = (u_1, u_3) \notin E(\sigma_2)$$

$$(v_2, v_4) \notin E(\sigma_1)$$

$$\text{||} (f(v_2), f(v_4)) = (u_2, u_3) \notin E(\sigma_2)$$

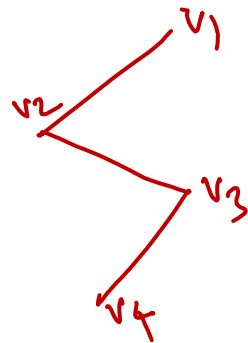
$$(v_1, v_3) \notin E(\sigma_1)$$

$$\text{||} (f(v_1), f(v_3)) = (u_1, u_4) \notin E(\sigma_2)$$

Thus, given G_1 and G_2 are isomorphic under vertex preserves condition

Method 2 [For f edge preserves condition]

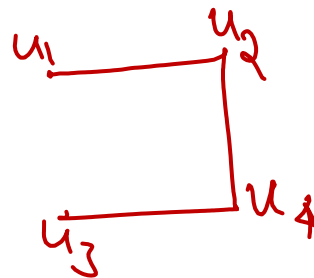
$$A_{G_1} = \begin{matrix} & v_1 & v_2 & v_3 & v_4 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$



Result
 $A_{G_1} = A_{G_2}$
 $\therefore G_1$ and G_2
 are
 isomorphic

$A_{G_2} \Rightarrow$ here rows and column of G_2 labeled from image of G_1

$$A_{G_2} = \begin{matrix} & u_1 & u_2 & u_3 & u_4 \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

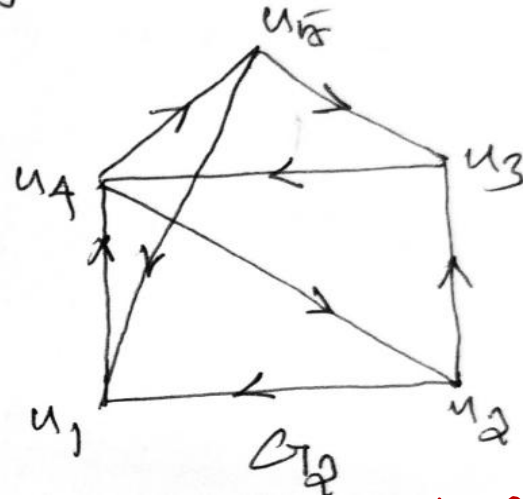
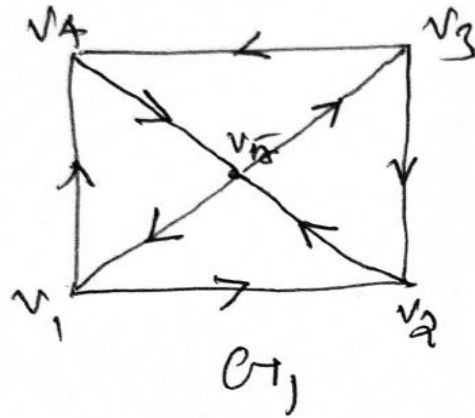


Note Isomorphism preserves the degree of vertices

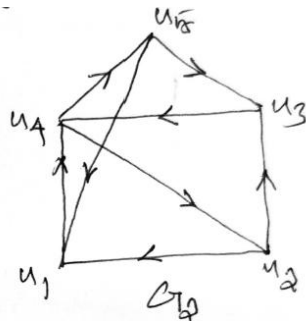
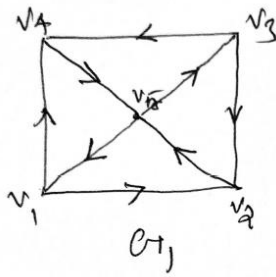
1. Let f be an isomorphism of the graph $G_1 = (V_1, E_1)$ to graph $G_2 = (V_2, E_2)$, let $v \in V_1$, then $\deg v = \deg f(v)$.

2. ✓ Two directed graphs are isomorphic if their respect underlying undirected graphs are isomorphic and are oriented the same.

Example show that the digraphs are isomorphic



Soln Given G_1 and G_2 having 5 vertices and 8 edges
Consider the indegree and outdegree of G_1 and G_2



G_1	indeg	out deg	G_2	indeg	outdeg
v_1	1	2	u_1	2	1
v_2	2	1	u_2	1	2
v_3	1	2	u_3	2	1
v_4	2	1	u_4	2	2
v_5	2	2	u_5	1	2

but we define a f as

$$f(v_1) = u_2$$

$$f(v_2) = u_1$$

$$f(v_3) = u_3$$

$$f(v_4) = u_4$$

$$f(v_5) = u_5$$

(clearly f is 1-1)

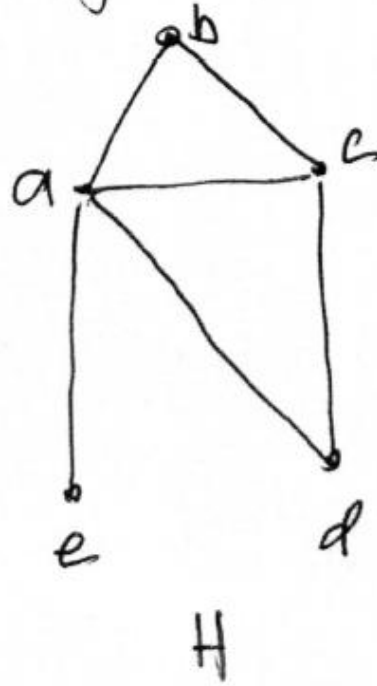
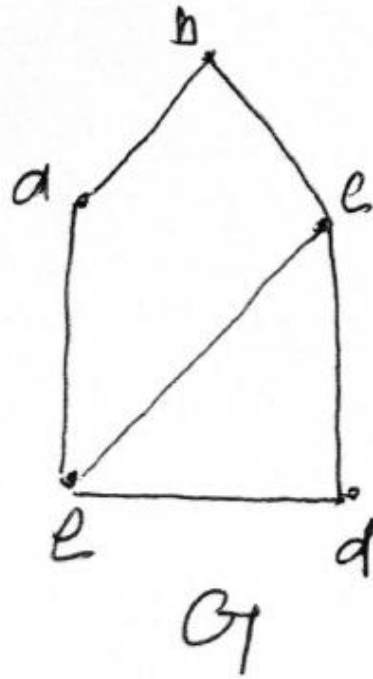
and onto

$\therefore G_1$ and G_2 are

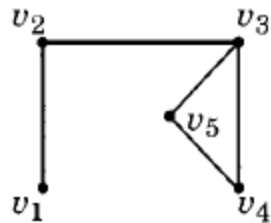
isomorphic

To check $A_{G_1} = A_{G_2}$ ✓

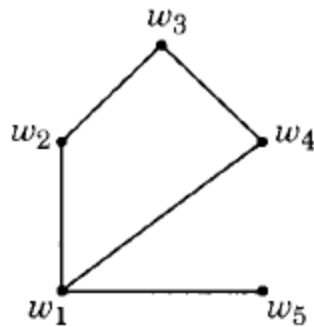
show that following graphs
are not isomorphic



Check the isomorphism between the graphs G_1 and G_2 .



Graph G_1



Graph G_2